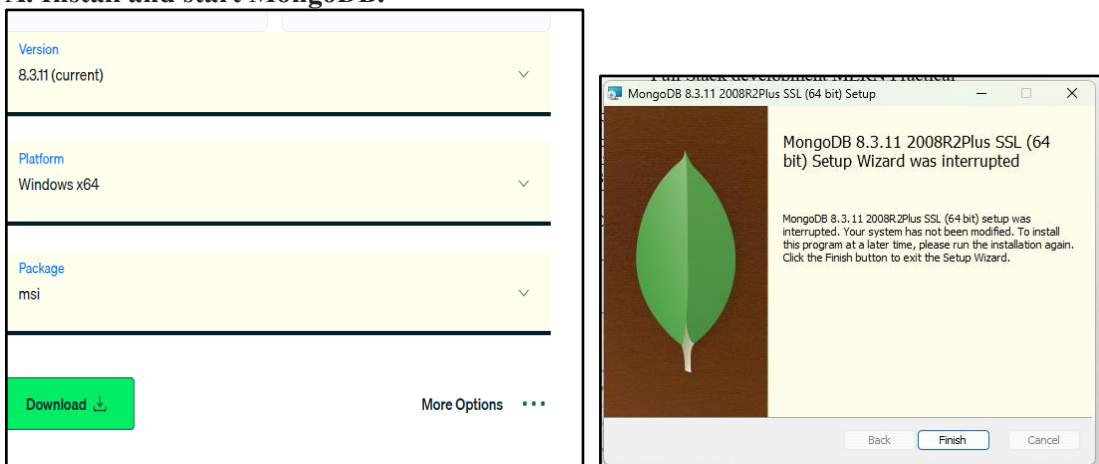


Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Full Stack development MERN Practical
Practical- 7

Roll No.: T025	Name: Afsha Amjad Mujawar
Class: TYIT	Batch:1
Date of Assignment: 19-09-2026	Date/Time of Submission: 19-09-2026

Aim: MongoDB Basics

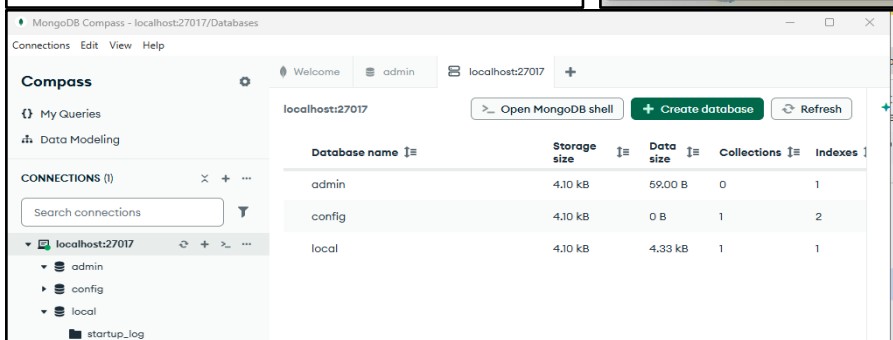
A. Install and start MongoDB.



The first screenshot shows the MongoDB download page with the following details:

- Version: 8.3.11 (current)
- Platform: Windows x64
- Package: msi
- Buttons: Download, More Options

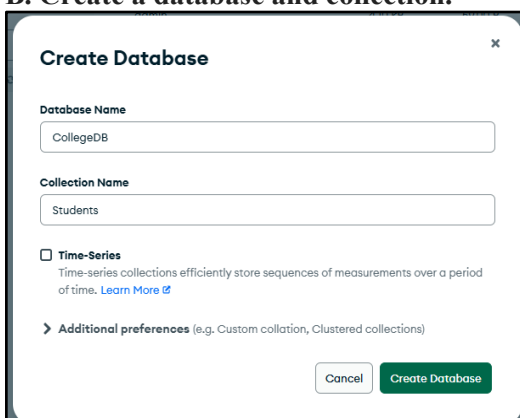
The second screenshot shows the MongoDB 8.3.11 2008R2Plus SSL (64 bit) Setup Wizard. The message states: "MongoDB 8.3.11 2008R2Plus SSL (64 bit) Setup Wizard was interrupted. MongoDB 8.3.11 2008R2Plus SSL (64 bit) setup was interrupted. Your system has not been modified. To install this program at a later time, please run the installation again. Click the Finish button to exit the Setup Wizard." Buttons: Back, Finish, Cancel.



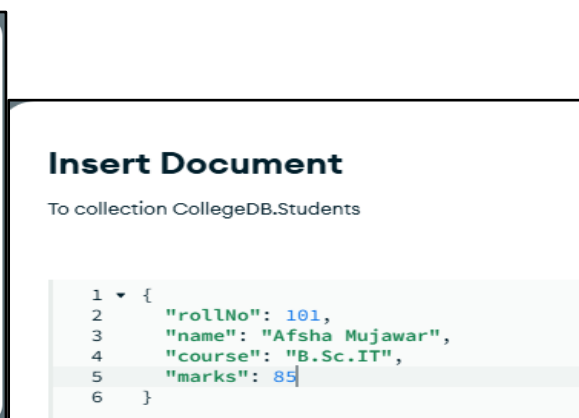
The third screenshot shows the MongoDB Compass interface. The left sidebar shows the 'Connections' list with 'localhost:27017' selected. The main panel shows the 'admin' database selected, with a table listing databases:

Database name	Storage size	Data size	Collections	Indexes
admin	4.10 kB	59.00 B	0	1
config	4.10 kB	0 B	1	2
local	4.10 kB	4.33 kB	1	1

B. Create a database and collection.



The first screenshot shows the 'Create Database' dialog box. The 'Database Name' field contains 'CollegeDB' and the 'Collection Name' field contains 'Students'. There is a checkbox for 'Time-Series' which is unchecked. At the bottom, there are 'Cancel' and 'Create Database' buttons.



The second screenshot shows the 'Insert Document' dialog box. The title is 'Insert Document' and the subtitle is 'To collection CollegeDB.Students'. The document to be inserted is shown in a JSON format:

```

1 {
2   "rollNo": 101,
3   "name": "Afsha Mujawar",
4   "course": "B.Sc.IT",
5   "marks": 85
6 }
```

C. Insert records in MongoDB.

```
> use("CollegeDB")
< switched to db CollegeDB
> db["Students"].find()
< {
  _id: ObjectId('6aae2a6d60aba6af85053765'),
  rollNo: 101,
  name: 'Afsha Mujawar',
  course: 'B.Sc.IT',
  marks: 85
}
```

- Insert single record
- insertOne() → inserts one document

```
> db.students.insertOne({})
< {
  acknowledged: true,
  insertedId: ObjectId('6aae2c0cd24ba64ca70fe89b')
}
```

- insertMany() → inserts multiple documents

```
> db.students.insertMany([ { rollNo:102,name:"Aashu",course:"B.Sc.IT",marks:92},{ ro
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6aae2dead24ba64ca70fe89c'),
    '1': ObjectId('6aae2dead24ba64ca70fe89d'),
    '2': ObjectId('6aae2dead24ba64ca70fe89e')
  }
}
```

D. View records from the collection.

- db.students.find() - View all records. This displays all student documents.
- db.students.find().pretty() - View records in a readable format

```
> db.students.find()
< {
  _id: ObjectId('6aae2c0cd24ba64ca70fe89b')
}
{
  _id: ObjectId('6aae2dead24ba64ca70fe89c'),
  rollNo: 102,
  name: 'Aashu',
  course: 'B.Sc.IT',
  marks: 92
}
{
  _id: ObjectId('6aae2dead24ba64ca70fe89d'),
  rollNo: 103,
  name: 'Asmitha',
  course: 'B.Sc.IT',
  marks: 78
}
{
  _id: ObjectId('6aae2dead24ba64ca70fe89e'),
  rollNo: 104,
  name: 'Radikha',
  course: 'B.Sc.IT',
  marks: 88
}
```

- Find a particular student-Find student with marks 92

```
> db.students.find({rollNo:102})
< {
  _id: ObjectId('6aae2dead24ba64ca70fe89c'),
  rollNo: 102,
  name: 'Aashu',
  course: 'B.Sc.IT',
  marks: 92
}
```

E. Update and delete records.

- Delete - Delete the student with roll number 104.
- deleteOne - deletes only 1 document
- deleteMany - deletes all matching documents

```
> db.students.deleteOne({ rollNo:104})
< {
  acknowledged: true,
  deletedCount: 1
}
```

- Update - Rahul's marks need to be changed from 85 to 90.
- **updateOne - updates only 1 document**

```
> db.students.updateOne({rollNo:102},{ $set:{marks:90}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

- **updateMany - updates all matching documents**

```
> db.students.updateMany({course:"B.Sc.IT"},{$set:{course:"Information Technology"}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
```

Output:

```
{
  _id: ObjectId('6aae2dead24ba64ca70fe89c'),
  rollNo: 102,
  name: 'Aashu',
  course: 'Information Technology',
  marks: 90
}
```

```
> db.students.find()
< {
  _id: ObjectId('6aae2c0cd24ba64ca70fe89b')
}
{
  _id: ObjectId('6aae2dead24ba64ca70fe89c'),
  rollNo: 102,
  name: 'Aashu',
  course: 'B.Sc.IT',
  marks: 92
}
```

F Connect MongoDB with [Node.js](#).

Code:

```
const mongoose = require("mongoose");
mongoose.connect("mongodb://127.0.0.1:27017/CollegeDB")
  .then(() => {
    console.log("MongoDB Connected Successfully");
  })
  .catch((error) => {
    console.log("MongoDB Connection Failed");
    console.log(error);
  });
```

```
PS C:\Users\mv1ui\Desktop\Afsha 25\mongo db> node server.js
MongoDB Connected Successfully
```

Code:

```
const mongoose = require("mongoose");
mongoose.connect("mongodb://127.0.0.1:27017/CollegeDB")
  .then(async () => {
    console.log("MongoDB Connected Successfully");
```

```
const students = await mongoose.connection.db
  .collection("students")
  .find()
  .toArray();
console.log(students);
})
.catch((error) => {
  console.log("MongoDB Connection Failed");
  console.log(error);
});
```

```
PS C:\Users\mvlui\Desktop\Afsha 25\mongo db> node server.js
MongoDB Connected Successfully
[
  { _id: new ObjectId('6aae2c0cd24ba64ca70fe89b') },
  {
    _id: new ObjectId('6aae2dead24ba64ca70fe89c'),
    rollNo: 102,
    name: 'Aashu',
    course: 'Information Technology',
    marks: 90
  },
  {
    _id: new ObjectId('6aae2dead24ba64ca70fe89d'),
    rollNo: 103,
    name: 'Asmitha',
    course: 'Information Technology',
    marks: 78
  }
]
```